Eunoia Junior College
9729 H2 Chemistry
2017 Paper 2 Suggested Solution

1 (a) (i) $\mathbf{A}^{+}(g) \rightarrow \mathbf{A}^{2+}(g) + e^{-}$

(a) (ii) Successive ionisation energy generally increases with the number of electrons removed as each subsequent electron is removed from an increasingly positive ion wherein the attraction becomes stronger.

The 7th electron is removed from the outermost (valence) shell, while the 8th electron is from the penultimate shell that is very much nearer to the nucleus, and hence experiences much stronger attractive forces. As a result, it requires much more energy to remove the 8th electron as seen in the significant jump in values from the 7th to the 8th ionisation energy.

(a) (iii) element **A** is **chlorine**.
$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

(b) (i) Analogous to the Group 1 metals, hydrogen possesses one valence electron in its sole 1s orbital.

(b) (ii) Unlike the Group 1 metals which have giant metallic structure, hydrogen exists as simple molecular diatomic H_2 molecules. As a result, the Group 1 metals are metals with reasonably high melting points, while hydrogen is a gas with very low melting point.

2 (a) $$[\text{HCO}_2\text{H}] = \frac{0.0100 \text{ mol}}{\frac{250}{1000} \text{ dm}^3} = 0.0400 \text{ mol dm}^{-3}$$

$$[\text{H}^+] \approx \sqrt{K_a \times [\text{HCO}_2\text{H}]} = \sqrt{1.60 \times 10^{-4} \times 0.0400} = 2.530 \times 10^{-3} \text{ mol dm}^{-3}$$

$$\text{pH} = -\lg[\text{H}^+] = \underline{\mathbf{2.60}}$$

(b) Using the K_a of HCO_2H, $K_a = \dfrac{[\text{H}^+][\text{HCO}_2^-]}{[\text{HCO}_2\text{H}]}$.

Due to suppression of the dissociation of HCO_2H by HCO_2^- from HCO_2Na,
$[\text{HCO}_2^-] \approx [\text{HCO}_2\text{Na}]_{\text{initial}}$ and $[\text{HCO}_2\text{H}] \approx [\text{HCO}_2\text{H}]_{\text{initial}}$

At pH 3.70, $[\text{H}^+] = 10^{-3.70} = 1.995 \times 10^{-4}$ mol dm^{-3}

$$1.60 \times 10^{-4} = \frac{(1.995 \times 10^{-4})\left(\frac{100}{100+150}[\text{HCO}_2\text{Na}]\right)}{\left(\frac{150}{100+150} \times 0.0100\right)}$$

$$[\text{HCO}_2\text{Na}] = \frac{250}{100} \times \frac{1.60 \times 10^{-4} \times \left(\frac{150}{100+150} \times 0.0100\right)}{1.995 \times 10^{-4}}$$
$$= \underline{\mathbf{0.0120 \text{ mol dm}^{-3}}}$$

or Using Henderson's equation,

$$\text{pH} = \text{p}K_\text{a} + \lg\frac{[\text{HCO}_2^-]}{[\text{HCO}_2\text{H}]}$$

$$3.70 = -\lg(1.60\times10^{-4}) + \lg\left(\frac{\frac{100}{100+150}[\text{HCO}_2\text{Na}]}{\frac{150}{100+150}(0.0100)}\right)$$

$$3.70 = 3.796 + \lg\left(\frac{2}{3}\times\frac{[\text{HCO}_2\text{Na}]}{0.0100}\right)$$

$$[\text{HCO}_2\text{Na}] = \underline{\mathbf{0.0120\ mol\ dm^{-3}}}$$

(c) (i) The pH of blood will **decrease**.

(c) (ii) The conjugate base, HCO_3^-, will react with H^+ from the lactic acid:

$$\text{HCO}_3^-(\text{aq}) + \text{H}^+(\text{aq}) \rightarrow \text{H}_2\text{CO}_3(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{aq}) + \text{CO}_2(\text{aq})$$

removing the lactic acid produced in the muscle, thus maintaining the pH of blood.

3 (a) (i) In 100 g of **D**,

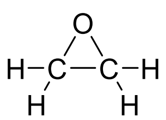
	C	H	O
Mass / g	54.5	9.10	36.4
Amt / mol	$\frac{54.5}{12.0} = 4.54$	$\frac{9.10}{1.0} = 9.10$	$\frac{36.4}{16} = 2.275$
Simplest mole ratio	$\frac{4.54}{2.275} = 2$	$\frac{9.10}{2.275} = 4$	$\frac{2.275}{2.275} = 1$

Hence the empirical formula of **D** is C_2H_4O.

Let the molecular formula of **D** be $C_{2n}H_{4n}O_n$.

$$\text{Molar mass of } \mathbf{D} = 44$$
$$2n(12.0) + 4n(1.0) + n(16.0) = 44$$
$$44n = 44$$
$$n = 1$$

Hence, the molecular formula is C_2H_4O.

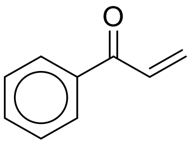
(a) (ii) The third possible isomer of **D** is ethene oxide,  .

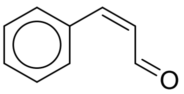
(b) Table 3.1

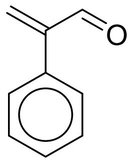
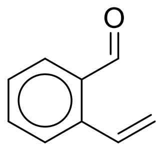
reagent	observation
aqueous bromine	**Orange aqueous bromine is decolourised**
aqueous sodium carbonate	**No CO_2 gas evolved**

(c) The very small K_c value indicates that the position of equilibrium **lies very much to the left**, with the major species being **ethanal**.

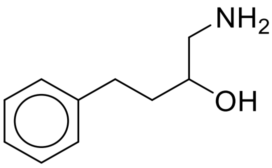
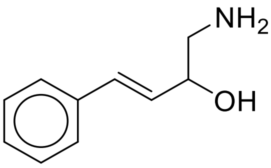
(d) $CH_3CHO(aq) + 3I_2(aq) + 4OH^-(aq) \rightarrow CHI_3(s) + HCO_2^-(aq) + 3I^-(aq) + 3H_2O(l)$

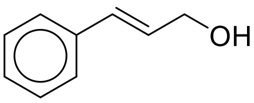
4 (a) isomer **E** is , **constitutional** (functional group) **isomer**.

isomer **F** is , ***cis-trans*** **isomer** (stereoisomer).

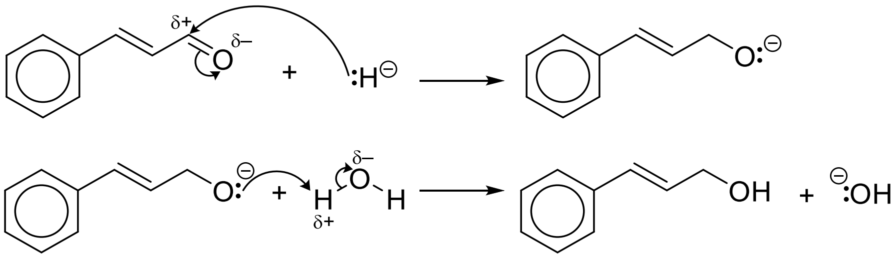
Other constitutional isomers are possible, *e.g.* , , *etc.*

(b) (i) HCN(aq), with trace of NaCN(aq), in the cold.

(b) (ii) **H** is , **I** is .

(c) (i) **J** is .

(c) (ii) The mechanism is



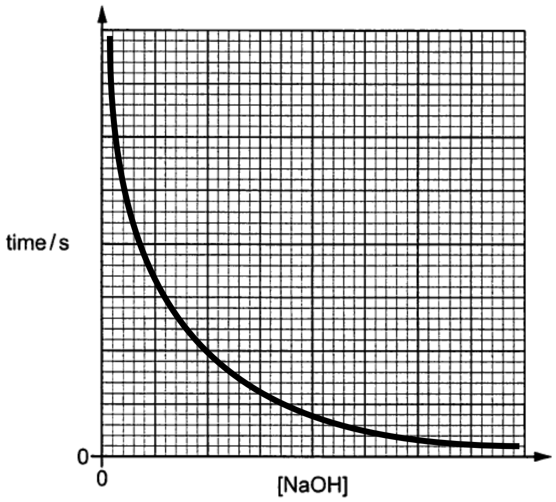
(c) (iii) As Al is more electropositive than B, the $\overset{\delta+}{\text{A}l}\text{—}\overset{\delta-}{\text{H}}$ bond is more polar and the H in AlH_4^- will have a larger partial negative charge (δ–) and hence **more nucleophilic** than that in BH_4^-.

(d) (i) Repeat the experiment using 20 cm^3 of cinnamaldehyde solution, x **cm^3 of 2 mol dm^{-3} sodium hydroxide and (12 – x) cm^3 of deionised water**. Record the **time taken (*t* seconds)** for the precipitate to appear when 15 drops of propanone is added to the mixture, keeping the **stirring consistent** with the first experiment.

The concentration of NaOH in the new experiment is $\frac{x}{12}$ times that of the first experiment. Since the rate of the reaction is **inversely proportional** to the time taken for the precipitate to appear, by determining how $\frac{1}{t}$ compares with $\frac{1}{50}$, one would be able to determine the effect on the rate of the reaction.

(d) (ii) Since rate $\propto [\text{NaOH}]$, and rate $\propto \frac{1}{\text{time}}$, so $[\text{NaOH}] \propto \frac{1}{\text{time}}$ or time $= \frac{k}{[\text{NaOH}]}$.

Hence the graph of time against $[\text{NaOH}]$ will be a hyperbola with the two axes as asymptotes.



5 (a) First, obtain the $E^{\ominus}_{\text{cell}}$ for the reaction:

$$E^{\ominus}_{\text{cell}} = E^{\ominus}_{\text{reduction}} - E^{\ominus}_{\text{oxidation}} = E^{\ominus}\left(S_2O_8^{2-}\middle|SO_4^{2-}\right) - E^{\ominus}\left(I_2\middle|I^-\right)$$

$$= +2.01 - (+0.54) = +1.47 \text{ V}$$

For the reaction given, 2 moles of electrons are involved per mole of $S_2O_8^{2-}$.

Using $\Delta G^{\ominus} = -nFE^{\ominus}$, $\Delta G^{\ominus} = -2 \times 96500 \times 1.47$

$$= -283710 \text{ J mol}^{-1}$$

$$\approx \underline{\mathbf{-284\ kJ\ mol^{-1}}}$$

Hence the reaction is **spontaneous**.

(b) When silver nitrate crystals are added to the I_2/I^- half-cell, AgI will be precipitated leading to a decrease in the concentration of I^- in the half-cell. As a result, $E^{\ominus}\left(I_2\middle|I^-\right)$ will **increase** since by Le Chaterlier's Principle, reduction of I_2 to I^- will be favoured to counter the decrease in I^- concentration. Since the standard cell potential, $E^{\ominus}_{\text{cell}} = E^{\ominus}\left(S_2O_8^{2-}\middle|SO_4^{2-}\right) - E^{\ominus}\left(I_2\middle|I^-\right)$, $E^{\ominus}_{\text{cell}}$ will **decrease**.

(c) (i) Since the reactants, $S_2O_8^{2-}$ and I^-, and the Fe^{2+} are all in the **same aqueous phase**, and the introduction of a **small amount** of Fe^{2+} helps to **increase the rate** of reaction, hence the iron(II) ions are described as a homogeneous catalyst.

(c) (ii) Iron exhibits **variable oxidation states**, allowing it to function as an electron carrier between the two negatively charged reactants, $S_2O_8^{2-}$ and I^-:

$$2Fe^{2+}(aq) + S_2O_8^{2-}(aq) \rightarrow 2Fe^{3+}(aq) + 2SO_4^{2-}(aq)$$
$$2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$$

(d) The complex in solid hydrated iron(III), $\left[Fe(H_2O)_6\right]^{3+}$, appears violet meaning that yellow light is absorbed. In aqueous iron(III), one of the H_2O ligand is replaced by a OH^-. The negatively charged OH^- will interact more strongly with the 3d orbitals of Fe^{3+} compared to H_2O, resulting in a **larger d-orbital splitting**. Hence, light of **shorter wavelength** (higher energy blue-violet region) will be absorbed, giving rise to the complementary yellow or brown colour observed.

Between iron(II) and iron(III), the more positive Fe^{3+} will **interact more strongly** with the ligands compared to Fe^{2+}, hence a **larger d-orbital splitting** is also expected in $\left[Fe(H_2O)_6\right]^{3+}$ compared to $\left[Fe(H_2O)_6\right]^{2+}$. $\left[Fe(H_2O)_6\right]^{2+}$ thus absorbs light of a **longer wavelength** (lower energy orange region), giving rise to the complementary green colour observed.

6 (a) (i) Benzoic acid can be obtained in a high state of purity, is not volatile and is stable (does not decompose and does not absorb moisture). In addition, benzoic acid readily undergoes complete combustion to give carbon dioxide and water only, and its enthalpy of combustion is known with great accuracy.

(a) (ii) By adjusting the temperature of the controlled temperature water jacket to be the same as the temperature of the calorimeter.

(b) (i) Some of the benzoic acid **did not burn completely** to give CO_2, instead giving the black coating of carbon (soot), hence less heat is given out.

(b) (ii) The chemist could have increased the pressure and hence amount of oxygen gas in the bomb, to ensure complete combustion.

(b) (iii) Molar mass of benzoic acid $= (12.0 \times 7) + (1.0 \times 6) + (16.0 \times 2) = 122 \text{ g mol}^{-1}$

Amount of benzoic acid used $= \dfrac{6.10 \text{ g}}{122 \text{ g mol}^{-1}} = 0.0500 \text{ mol}$

Amount of heat evolved $= 3230 \text{ kJ mol}^{-1} \times 0.0500 \text{ mol} = 161.5 \text{ kJ}$

Average temperature rise $= \dfrac{(55.3 - 25.0) + (54.7 - 25.0)}{2} = 30.0 \text{ °C}$

Heat capacity of calorimeter $= \dfrac{161.5 \text{ kJ}}{30.0 \text{ °C}} = \underline{\mathbf{5.38 \text{ kJ °C}^{-1}}}$

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(c) (i) $C_7H_6O_2(s) + \dfrac{15}{2}O_2(g) \rightarrow 7CO_2(g) + 3H_2O(l)$

(c) (ii) This is because the volume is fixed inside the bomb, due to changes in the pressure of the gases inside the bomb, work is done in addition to the heat evolved.

Or the pressure of oxygen gas in the bomb is much higher than the 1 bar for standard condition, hence giving rise to the discrepancy in the heat changes.

7 (a) (i) Enthalpy change $= 5 \times -118 = \underline{\mathbf{-590\ kJ\,mol^{-1}}}$

(a) (ii) Having a **less exothermic** enthalpy of hydrogenation, naphthalene is (590 – 335 =) **255 kJ mol^{-1} more stable** than the hypothetical molecule with 5 C=C bonds.

(a) (iii) Each of the 10 carbon atoms in naphthalene is **sp^2 hybridised**, with one singly filled 2p orbital perpendicular to the plane of the molecule. A **C–C σ bond** is formed with neighbouring C atoms *via* **sp^2–sp^2 overlap**, while all 10 p orbitals with one electron each overlap sideway leading to a **delocalised electron cloud of 10 π electrons** covering all 10 C atoms. The bonds between the carbon atoms are $\text{C}\overline{\cdots}\text{C}$ bonds, in between a single C–C and double C=C bond.

(b) (i) Using the ideal gas equation,

$$pV = nRT$$

$$\left(101\times10^{3}\right)\left(125\times10^{-6}\right) = n_{H_2} \times 8.31\times\left(30+273\right)$$

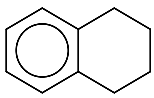
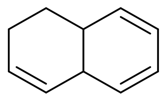
$$n_{H_2} = 5.014\times10^{-3}\ \text{mol}$$

$$n_{\text{naphthalene}} = \frac{0.32\ \text{g}}{\left(12.0\times10+1.0\times8\right)\ \text{g mol}^{-1}} = 2.50\times10^{-3}\ \text{mol}$$

$$n_{\text{naphthalene}} : n_{H_2} = 2.50\times10^{-3} : 5.014\times10^{-3} = 1:2$$

Hence, each mole of naphthalene, $C_{10}H_8$, reacts with 2 moles of H_2, giving $C_{10}H_{12}$:

$$C_{10}H_8 + 2H_2 \rightarrow C_{10}H_{12}$$

(b) (ii) Compound **K** is , compound **L** is  (there are other structures possible for **L** so long as there are three double bonds)

(b) (iii) **K** is more likely to be formed since the resonance stabilised benzene ring remains intact, and hence will be more stable than **L** with three C=C double bonds.

(b) (iv) For complete hydrogenation of naphthalene, **excess hydrogen gas** should be used with **nickel catalyst** under **high temperature** and **high pressure**.